

Discussion of “Ranking Algorithms and Equilibrium Prices” by Huang, Korganbekova, and Korganbekova

Marketing-Industrial Organization Conference

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Adam Harris

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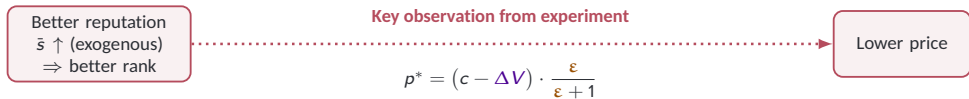
Comments:

1. Strengthening argument for rank $\nrightarrow$ demand elasticity
2. Exploring heterogeneity

Overview

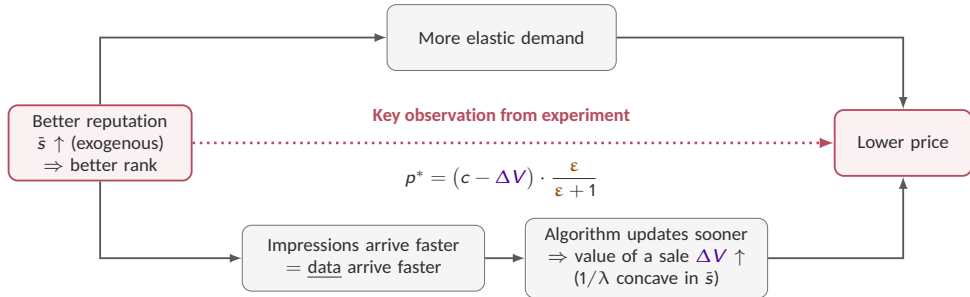


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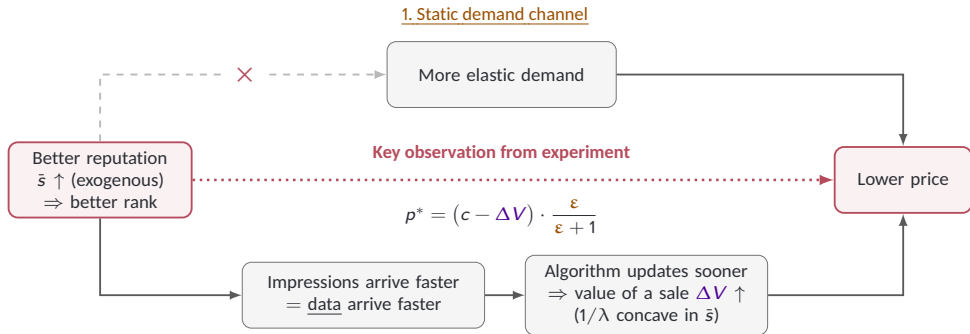
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1. Static demand channel



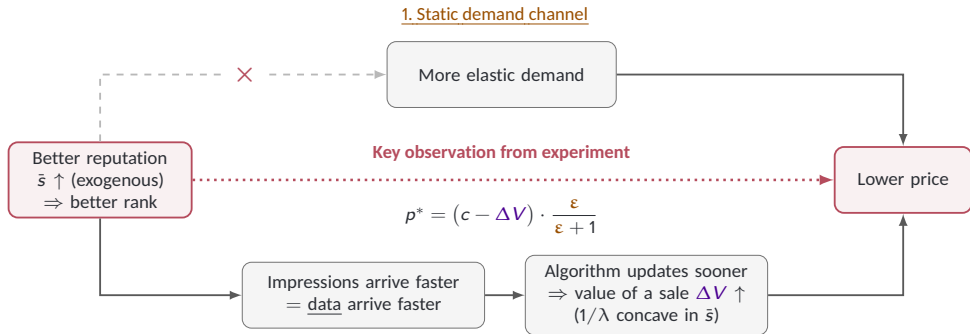
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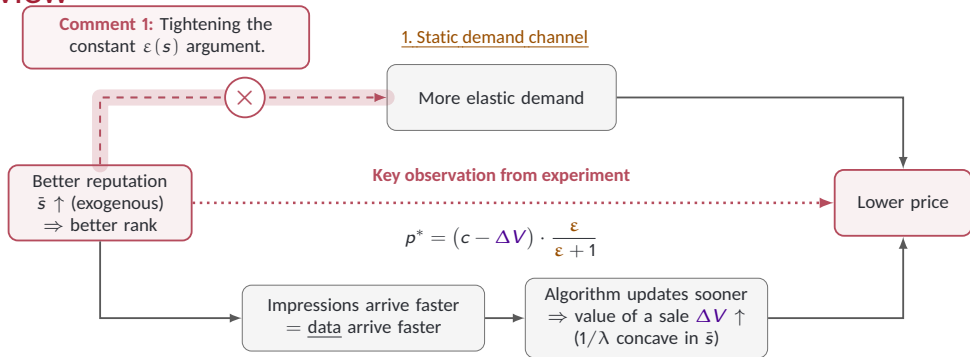


2. Dynamic incentive: invest in future reputation

Reputation is easier to maintain than to rebuild

- Traffic is the reward and the data the algorithm learns from.
- Better reputation raises the **return** to investing and the **capacity** to invest.
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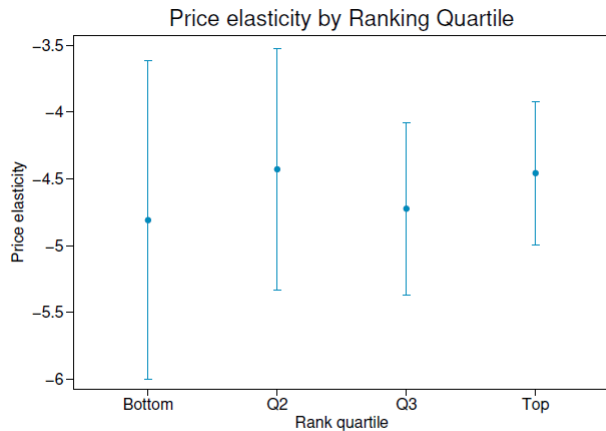
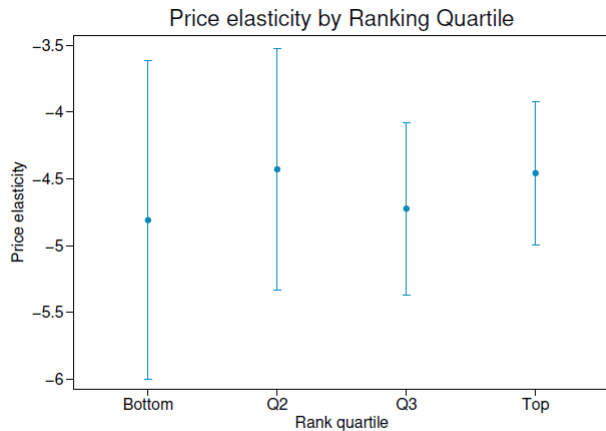


Figure 6: Estimated Price Elasticity by Rank Quartile

Comment 1: Demand elasticity



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- If experimental rank improvement makes $|\epsilon| = 4.5 \rightarrow 4.9$, that's enough to rationalize 2.3% decrease in prices.

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Two concrete suggestions

1. A more parsimonious functional form, e.g.,

$$\mathbb{E} [q_{jt} \mid p_{jt}, \mu_j, \delta_{gt}] = \exp (\alpha_1 \log (p_{jt}) + \alpha_2 \log (p_{jt}) \times \log (\text{rank}_{jt}) + \mu_j + \delta_{gt})$$

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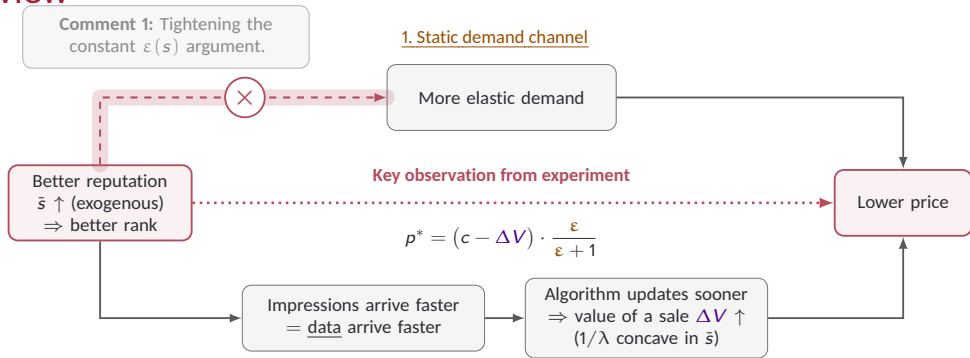
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2. Instead, run tests of the form: $H_0 : \alpha_2 \geq A$.
 - Then ask, what is the largest value A such that you fail to reject? Call this $A_{\text{worst case}}$.
 - If $\alpha_2 = A_{\text{worst case}}$, what proportion of the 2.3% price decrease would the demand elasticity channel explain?

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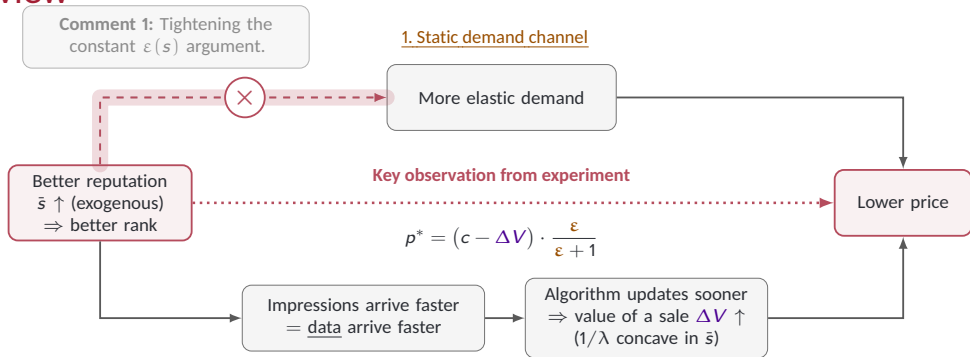


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“Reputation easier to maintain than to rebuild.” But does everyone want to maintain?

- Positive price responses for some product clusters: Which clusters are these?

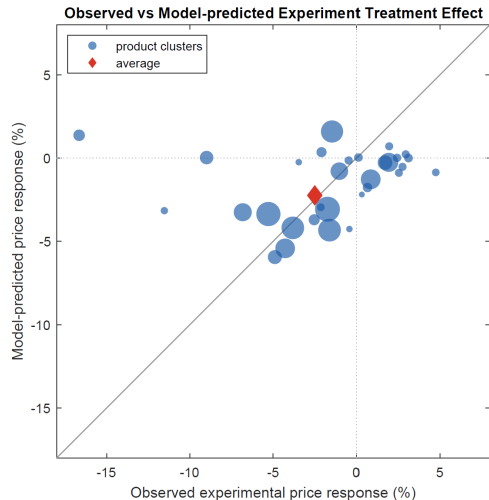


Figure B3: Model validation by cluster: predicted and observed price responses

Closing thoughts

- Clean experiment, and the mechanism is the right one.
- Two suggestions:
 - Demand elasticity: How much of the price response can a gradient in ε deliver?
 - Heterogeneity: For *which* products (clusters) does the story hold?

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Structural model:

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 - $F(\hat{s}_{jt-1})$ is a *descendant of past prices* — the investment mechanism itself.
 - Grouped FEs (Bonhomme, Lamadon, and Manresa (2022)) likely absorb ξ , so I'm not actually too worried about this in the structural model.